Base, uncertainty treatment and transition evidence across S1–S5

This audit separates search-layer membership from substantive uncertainty treatment and full-text evidence of transfer. It recomputes archived data; it does not rerun searches or introduce new screening decisions.

Stage	Base	U layer	Base ∩ U	% Base	Base outside U
S1	878	256	256	29.2%	622
S2	246	44	44	17.9%	202
S3	165	49	48	29.1%	117
S4	3098	543	543	17.5%	2555
S5	7593	1584	1584	20.9%	6009

S1–S3 are pre-screening merged corpora, including scope noise. S1 uncertainty membership includes reconstructed Scopus records and requires query verification. S3 has one U-only publication (DOI 10.1111/jiec.13604); it is not silently added to the Base denominator. S4–S5 use adjudicated canonical identifiers. Parsing both pipe and semicolon separators recovers memberships previously hidden by compound query cells. Different stage scopes and review histories prevent interpreting these percentages as comparative uncertainty-analysis prevalence.

The S4–S5 union contains 9,587 Base records and 1,934 U-layer records (20.2%). These are retrieval memberships, not final relevant studies. Among U-layer records, 392 clear candidates have archived title/abstract mechanism assessments; these assessments do not cover the complete Base or U sets.

Uncertainty treatment in assessed U candidates

Title/abstract classification	S4 query members	S5 query members
Mention / no demonstrated UQ	63/143 (44.1%)	112/331 (33.8%)
Sensitivity or scenarios only	43/143 (30.1%)	146/331 (44.1%)
Quantified uncertainty only	20/143 (14.0%)	30/331 (9.1%)
Explicit propagation (not necessarily cross-stage)	17/143 (11.9%)	43/331 (13.0%)

Stage groups overlap (82 shared records); their 143 and 331 assessments represent 392 distinct publications. Across the union, 144 only mention uncertainty or provide no demonstrated UQ, 156 use sensitivity/scenarios, 40 quantify uncertainty without an explicit propagation classification, and 52 are classified as explicit propagation. These historical title/abstract labels are not proof of propagation across an S boundary.

Cross-stage claims and full-text adjudication

Title/abstract S4–S5 claim	S4 query members	S5 query members
explicit	1/143 (0.7%)	3/331 (0.9%)
possible	37/143 (25.9%)	84/331 (25.4%)
not_demonstrated	105/143 (73.4%)	244/331 (73.7%)

Three union records carried explicit cross-stage claims in title/abstract screening. Full-text appraisal places Almeida et al. at S4→S5 with partial transfer; Aljawhari et al. and Caruso et al. concern other inputs→S5. Thus three screening claims must not be reported as three confirmed S4→S5 transfers.

Boundary	Assessed contexts	Partial transfer (P)	Quantified selected inputs (E)	Other codes
S1→S2	9	1	0	Q=1; N=4; R=1; S=2
S2→S3	6	3	2	N=1
S3→S4	0	0	0	—
S4→S5	1	1	0	—
Other inputs→S5	7	2	2	Q=1; S=2

Q: discussion/review; N: downstream transfer not demonstrated; R: inputs prepared for propagation; S: sensitivity/representation comparison. P and E must not be combined as proof of a complete uncertain chain. The 23 contexts come from 20 publications. A zero assessed-context denominator gives an undefined percentage, not zero prevalence.

PRISMA accounting and reproducibility

The selected set has 21 reports sought, one not retrieved, 20 assessed, zero full-text exclusions and 20 included. It is separate from the broad search denominators. Missing links between broad screening and focused selection cannot be reconstructed by subtraction. All source hashes, record-level memberships, category decisions and full-text context reasons accompany this report in CSV files. No unassessed record is labelled as lacking uncertainty analysis.

Population-wide uncertainty-treatment rates for S1–S5 remain unestablished. The reported rates are exact arithmetic for the specified archived cohorts, with the S1 provenance qualification retained. Wider substantive rates require systematic content review of the remaining records, including Base-only records.